



Enroll. No. _____

MARWADI UNIVERSITY

Faculty of Engineering and Technology

Information and Communication Technology

B.Tech.

SEM: V

MU FINAL EXAM (Open Book)

December 2024

Subject: - Machine Learning (01CT1519)

Date:- 26/12/2024

Total Marks:-60

Time: - 120 minutes

Question 1: Do as directed: [CO1,CO2] [12]

1. How can you cluster any given labelled data? Explain the process. [02]
2. Can you solve any given problem of regression using KNN? If yes, then how? If no, then Why? [02]
3. Write down the scenario for encoding the categorical values using: [02]
 1. One-Hot Encoding
 2. Label Encoding
4. Demonstrate non linear boundary of EX-OR gate. Show EX-OR gate realization with help of neural network and weights. [02]
5. You have a dataset with 50 features, but you suspect that many of them are irrelevant. What techniques can you use to select a subset of relevant features? [02]
6. How would you evaluate whether your model is suffering from high bias or high variance, and what techniques would you use to address these issues? [02]

Question 2: Solve the following [Attempt any two]: [CO1,CO2] [16]

1. Design the Neural Network with weights to solve the Boolean Equation: [08]

$$Y = (A \cdot \overline{B}) + (B \cdot C)$$

2. For the data set, construct the decision tree. [08]

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

3. Calculate the cost value of linear regression for first 2 epochs for the following dataset. [08]
 $X=[1,2,3,4,5]$
 $Y=[3,6,7,11,15]$
Slope of hypothesis (Θ_1) = 7
Constant/ intercept (Θ_0)=5
Learning rate (α) = 0.5

Question 3: Do as directed [Attempt any one]: [CO2,CO3] [08]

1. Determine the Principal Component for the given 2-Dimensional dataset. (1, 2), (2, 4), (3, 6). [08]
Represent the transformed points over the eigen axis.
2. Draw the clusters (with centroids) with the step-wise calculation to obtain the K-Means clustering [08]
over the data points. Consider number of clusters=3 and continue for 2 iterations
P1(1,3), P2(2,2), P3(5,8), P4(8,5), P5(3,9), P6(10,7), P7(3,3), P8(9,4), P9(3,7)

Question 4: Do as directed [Attempt any 2]: [CO4] [10]

1. You are provided with the following confusion matrix for a binary classification problem (2 classes: Positive and Negative). Design the classification report for this scenario [05]

	Predicted: Positive	Predicted: Negative
Actual: Positive	70	20
Actual: Negative	15	95

2. Consider the actual vs predicted value as follows: [05]

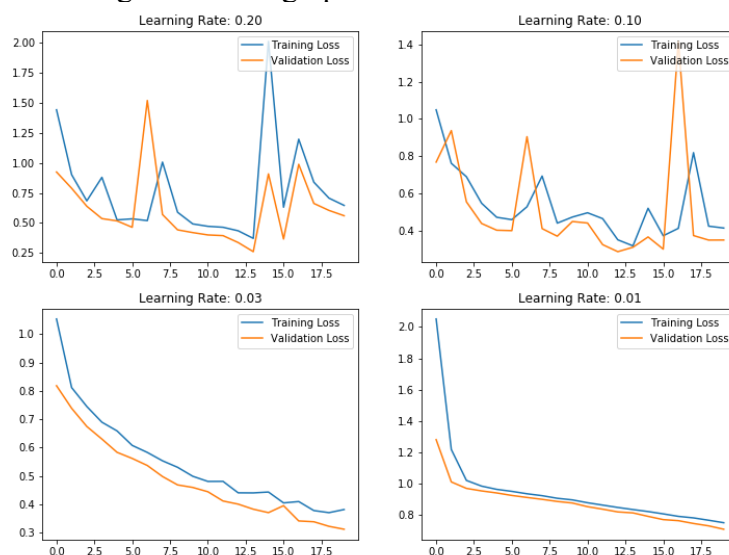
Actual Value	Predicted Value
5	4.5
7	6.8
10	9.7
12	13
15	14.2

Obtain following regression evaluation parameters:

MSE, RMSE, MAE, R-Square.

Also determine, is the predicted regression line is the best fit line?

3. Comment over the following evaluation graph: [05]

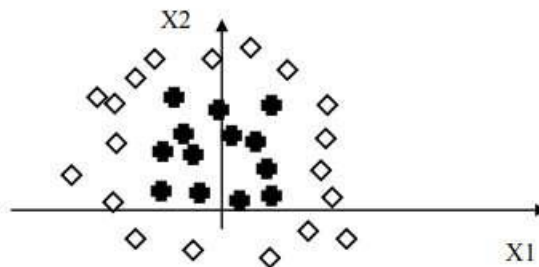


Question 5: Do as directed [Attempt any two]: [CO5]**[08]**

1. Given the dataset of temperature for all the months for last 100 years. Predict the temperature for next 1 year. Design the solution for the given problem statement. [04]
2. You are working on an image classification task where the input images are grayscale and have a size of 28x28 pixels. A convolutional layer in your CNN has the following configuration: [04]
 - Number of kernels (filters): 6
 - Kernel size: 3x3
 - Stride: 1
 - Padding: "same"
 1. Calculate the size of the convolved output feature map with the total number of feature maps obtained after this step
 2. If the stride is changed to 2 and padding is "valid," calculate the size of the convolved output feature map with the total number of feature maps
 3. Describe what happens to the feature maps if the kernel size is increased to 7x7.
 4. Explain how the pooling layer impacts the overall result of the convolution output?
3. Despite being designed to overcome the vanishing gradient problem, LSTM networks still have some limitations. What are the potential challenges you might face when training LSTMs on very long sequences, and how could you address them? [04]

Question 6: Do as directed [CO5]**[06]**

1. Suppose that we want to build a neural network that classifies two dimensional data (i.e., $X = [x_1, x_2]$) into two classes: diamonds and crosses. We have a set of training data that is plotted as follows: [06]



Draw a network that can solve this classification problem. Justify your choice of the number of nodes and the architecture. Draw the decision boundary that your network can find on the diagram

---Best of Luck---

Question	COs	Bloom's Level	POs
Q.1-1	CO1, CO2	Analyze	PO2
Q.1-2	CO1, CO2	Evaluate	PO1, PO2
Q.1-3	CO1, CO2	Apply	PO1
Q.1-4	CO1, CO2	Apply	PO1, PO3
Q.1-5	CO1, CO2	Analyze	PO2, PO4
Q.1-6	CO1, CO2	Apply	PO2, PO4
Q.2-1	CO1, CO2	Create	PO3, PO4
Q.2-2	CO1, CO2	Analyze	PO1, PO2
Q.2-3	CO1, CO2	Apply	PO1, PO2
Q.3-1	CO2, CO3	Analyze	PO1, PO4
Q.3-2	CO2, CO3	Apply	PO1, PO4
Q.4-1	CO4	Apply	PO1, PO2
Q.4-2	CO4	Evaluate	PO1, PO4
Q.4-3	CO4	Evaluate	PO2
Q.5-1	CO5	Create	PO3, PO5
Q.5-2	CO5	Analyze	PO1, PO5
Q.5-3	CO5	Analyze	PO2, PO5
Q.6-1	CO5	Create	PO3, PO5